Convex optimization - Project 4
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X is R.V. , $X \in \{1, \dots, n\}$, $\Theta \in \{1, \dots, m\}$, $P \in R^{n \times m}$ dist.

goal $\rightarrow$ design $T \in R^{m \times n}$, $D = TP \in R^{m \times m}$

D_{ij} $i \neq j$ is error D_{ii} accuracy , $t_{ik} = \text{Prob}(\hat{\Theta} = i \mid X = k)$
$i = 1, \dots, m$
$k = 1, \dots, n$

a)

$$\min \quad \max_{i \neq j} D_{ij}$$
$$\text{s.t.} \quad t_{ik} \geq 0 \quad i = 1, \dots, m, \; k = 1, \dots, n$$
$$\sum_i t_{ik} = 1 \quad i = 1, \dots, m, \; k = 1, \dots, n$$

b)

$$\max \quad \min_i D_{ii}$$
$$\text{s.t.} \quad t_{ik} \geq 0 \quad i = 1, \dots, m, \; k = 1, \dots, n$$
$$\sum_i t_{ik} = 1 \quad i = 1, \dots, m, \; k = 1, \dots, n$$

c)

$$\min \quad \frac{1}{m^2 - m} \sum_{i \neq j} D_{ij}$$
$$\text{s.t.} \quad t_{ik} \geq 0 \quad i = 1, \dots, m, \; k = 1, \dots, n$$
$$\sum_i t_{ik} = 1 \quad i = 1, \dots, m, \; k = 1, \dots, n$$

* Code is attached.